

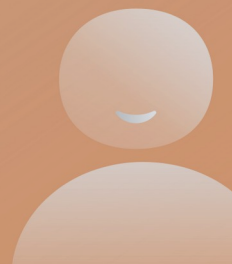


CM011 · MODULE 3 TECHNICAL REPORT PRESENTATION

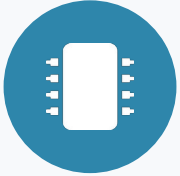
Semiconductor Fabrication Chemistry

Doping, Photolithography, and Etching in Silicon Chip Manufacturing

Marpel Mar Bangcaya · BS Computer Engineering · Special Topics in General Chemistry



What We'll Cover



Why Fabrication Chemistry Matters

From sand to silicon to system



Doping

Engineering silicon's electrical conductivity



The Environmental Cost

Chemicals, water, and greenhouse gases



Photolithography

Patterning a wafer with light and chemistry



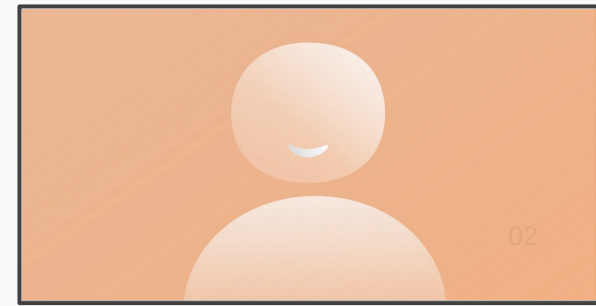
Etching

Carving the circuit into the wafer



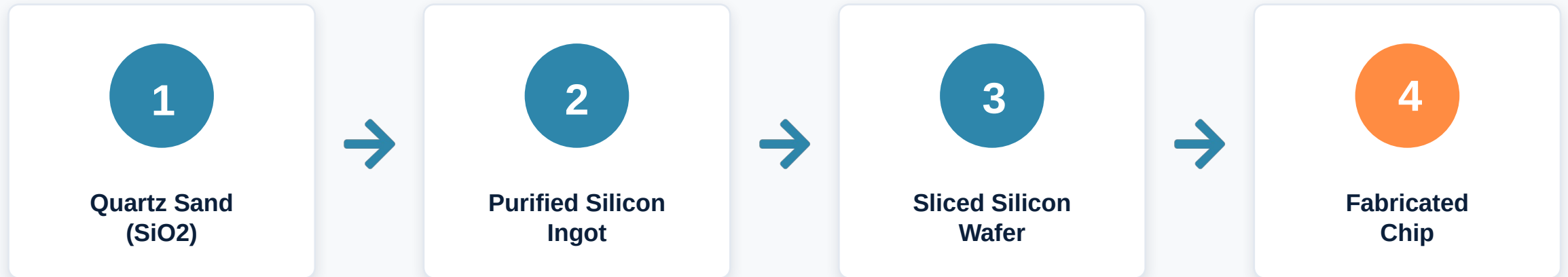
Why It Matters to You

Connecting chemistry to computer engineering

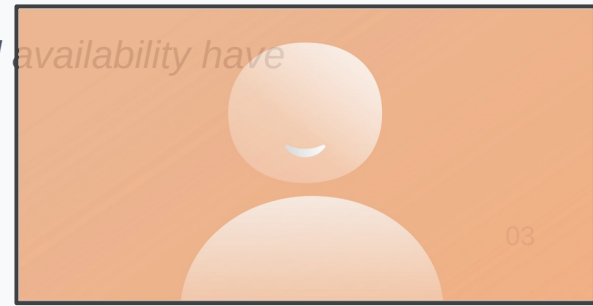


Every Chip Starts as Sand

Turning ordinary quartz sand into a working processor is fundamentally a chemistry problem, not a mechanical one.

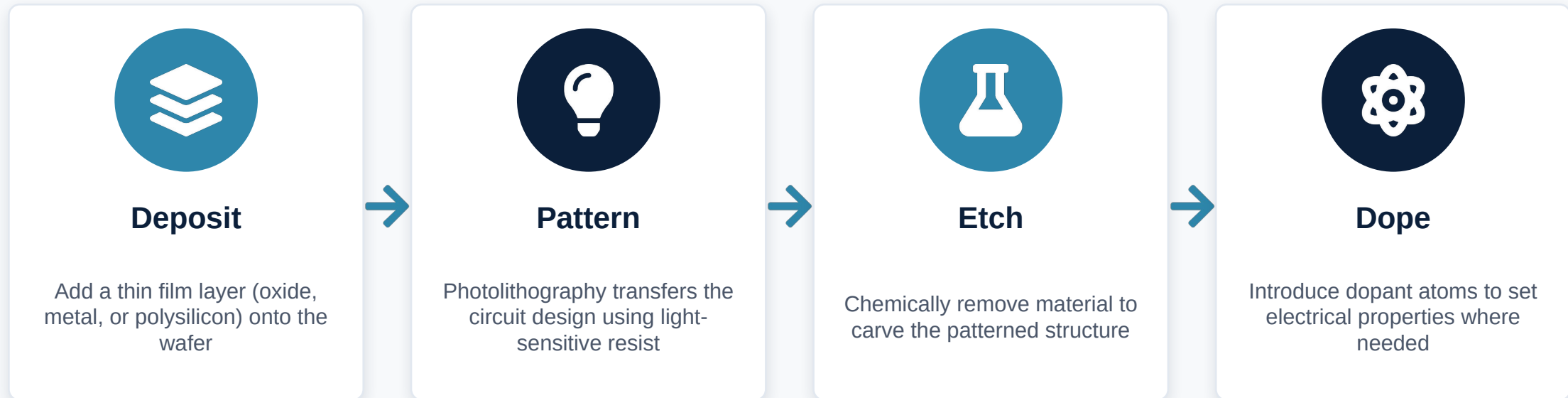


As a computer engineering student, understanding this chemistry explains why chip performance, cost, and availability have real physical and environmental limits.

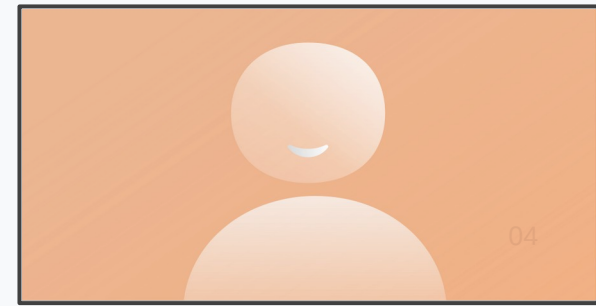


The Fabrication Cycle

These four chemical operations repeat dozens of times to build a single chip.



↳ Repeated 30–100+ times per wafer, building layer upon layer of transistors and interconnects

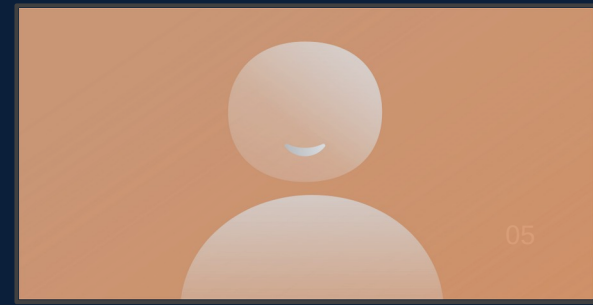




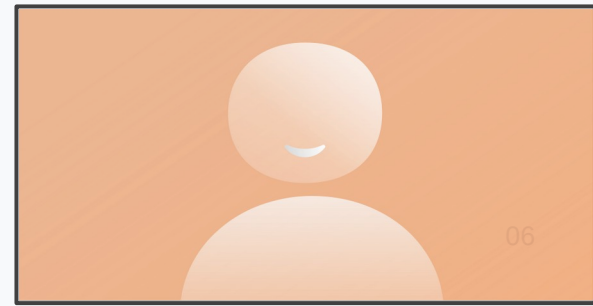
SECTION 1 OF 3

Photolithography

Chemically transferring a circuit pattern onto the wafer using light



Four Steps to a Pattern



Positive vs. Negative Resist

The same UV exposure produces opposite patterns, depending on the resist's chemistry.

POSITIVE

Positive Resist

Mechanism: Chain scission

UV light breaks the polymer's chains apart in the exposed areas, making them MORE soluble. The developer washes away the exposed regions, leaving the unexposed pattern behind.

NEGATIVE

Negative Resist

Mechanism: Cross-linking

UV light cross-links the polymer chains in the exposed areas, making them LESS soluble. The developer washes away the unexposed regions, leaving the exposed pattern behind.



Shorter Wavelength = Smaller Features

Rayleigh criterion: **$CD = k_1 \times (\lambda / NA)$** *(smallest printable feature)*

ArF Immersion DUV

$\lambda = 193 \text{ nm}$, $NA = 1.35$

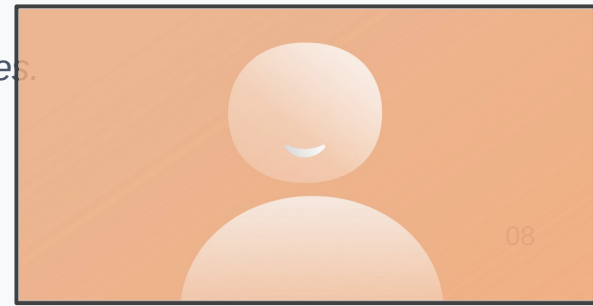
42.9 nm

EUV Lithography

$\lambda = 13.5 \text{ nm}$, $NA = 0.33$

12.3 nm

A ~15× shorter wavelength prints features about 3.5× smaller — why the industry moved to EUV for sub-10 nm nodes.

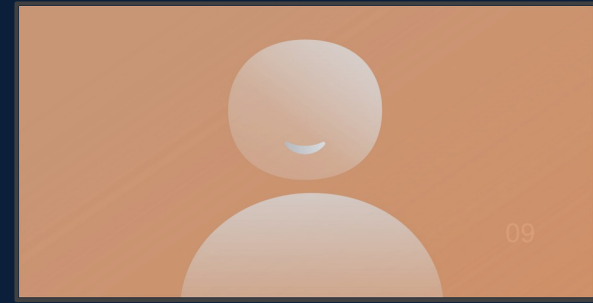




SECTION 2 OF 3

Doping

Chemically engineering silicon's electrical conductivity, atom by atom



Why Silicon Needs Impurities

Pure silicon is a poor conductor. Doping deliberately introduces trace impurity atoms into the silicon lattice to control exactly how many free charge carriers are available, and of which type.



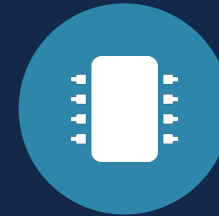
N-type

Phosphorus or arsenic donate extra electrons (negative carriers)

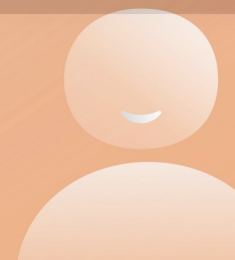


P-type

Boron accepts electrons, creating positive "holes" as carriers



Precisely placed n-type and p-type regions next to each other form the diodes and transistors that make up every logic gate on a chip.



Diffusion vs. Ion Implantation

Two chemically different ways to get dopant atoms into the wafer.



Diffusion

SOURCES

POCl_3 (n-type), BBr_3 (p-type), or gas-phase $\text{PH}_3/\text{B}_2\text{H}_6$

CONDITIONS

~900°C predeposition, driven in near 1200°C

CONTROL

Depth and dose are linked — harder to control independently

BEST FOR

Simpler, deeper, less critical junctions



Ion Implantation

SOURCES

Accelerated B^+ , P^+ , or As^+ ions from an ion beam

CONDITIONS

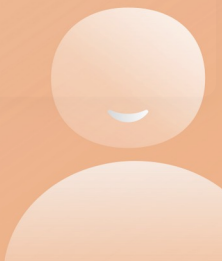
Room temperature; ~30 keV to several 100 keV, then ~1000°C anneal

CONTROL

Dose (beam current $\times$ time) and depth (voltage) set independently

BEST FOR

Shallow, precisely-doped junctions in modern nodes



Dose Is Set Directly by the Beam

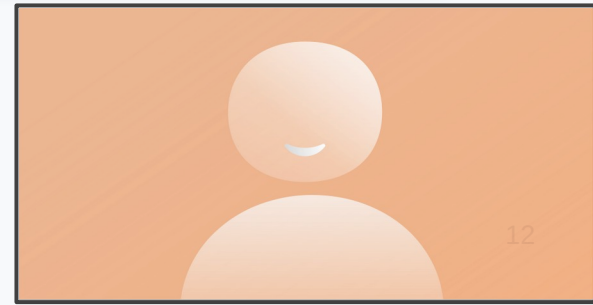
Implant dose: $Q = (I \times t) / (q \times A)$ *I = beam current, t = time, $q = 1.602 \times 10^{-19}$ C, A = area*

Example: $I = 1 \text{ mA}$, $t = 10 \text{ s}$, $A = 100 \text{ cm}^2$

6.2×10^{14}

ions/cm² — squarely within the 10^{13} – 10^{15} range fabs use for source/drain implants

Because dose and depth are set independently, implantation gives far tighter control than diffusion — why it replaced diffusion for critical junctions.

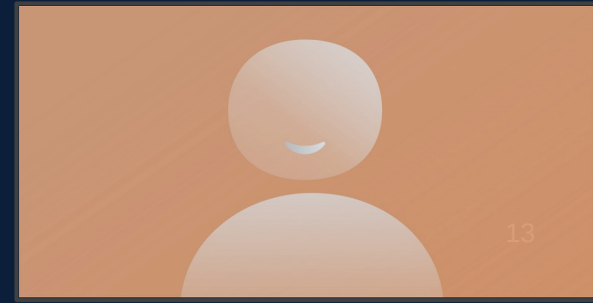




SECTION 3 OF 3

Etching

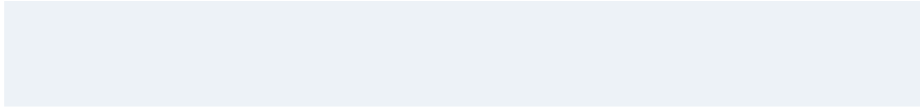
Chemically removing material to carve out the patterned structures



Wet Chemistry vs. Dry Plasma



Wet Etching



Isotropic — rounded sidewalls

CHEMICALS

HF, HNO₃, H₃PO₄, HCl; KOH, TMAH

SELECTIVITY

High — strong chemical selectivity

EQUIPMENT

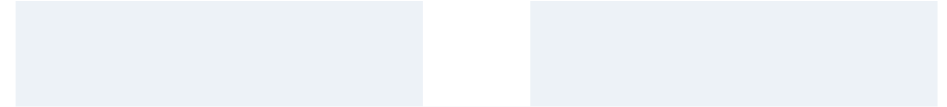
Simple, higher etch rate

USED FOR

Cleaning, oxide strip, less critical steps



Dry (Plasma) Etching



Anisotropic — near-vertical sidewalls

CHEMICALS

Fluorocarbon / chlorine / bromine plasmas

SELECTIVITY

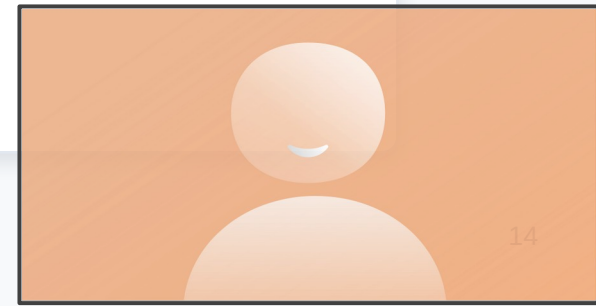
Lower — traded for geometric precision

EQUIPMENT

Complex vacuum/plasma systems

USED FOR

Nanometer-scale transistor structures



Chemistry Has a Footprint

Wet chemicals, PFAS, and high-GWP dry-etch gases make fabrication chemistry an environmental as well as a technical problem.



185M tons

CO₂-equivalent emissions from chip production annually



~50%

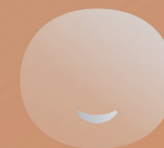
of all chemical use per wafer comes from wet processing



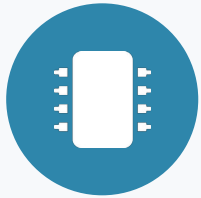
98% less

CF₄ consumption using pulsed "Transient Assisted Processing" (imec)

Fabs are responding with EUV dose reduction, pulsed low-consumption etching, ultrapure-water recycling, and PFAS-reduction programs.



Why This Matters to a Computer Engineer



Chip cost & availability

Resolution, junction, and etch limits set by this chemistry directly shape what chips exist, and at what price.



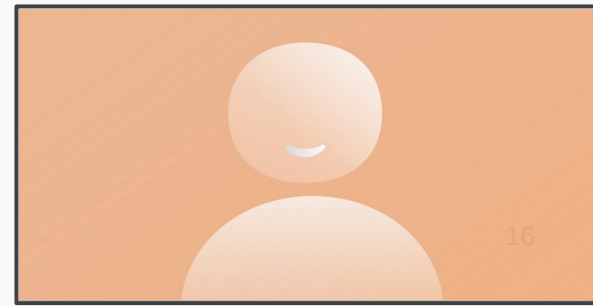
Performance ceilings

Smaller, better-controlled transistors (enabled by EUV, implantation, and dry etch) are what keep performance scaling.



Where chips can be made

Environmental and chemical-safety pressure increasingly determines fab location and capacity.



Three Things to Remember



Photolithography chemically prints the pattern

Shorter wavelengths (EUV) buy smaller features at the cost of process complexity.



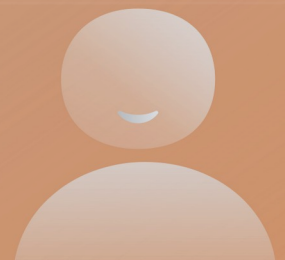
Doping chemically sets conductivity

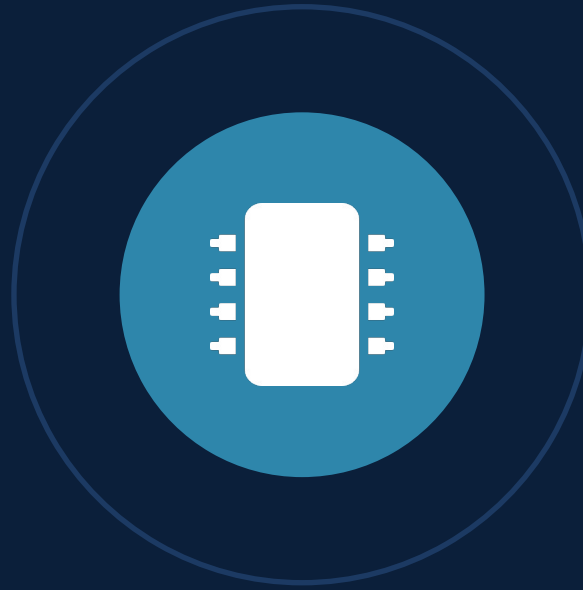
Ion implantation's independent dose/depth control has displaced diffusion for critical junctions.



Etching chemically carves the structure

Anisotropic dry plasma etching is what makes nanometer-scale transistors possible.





Thank You

